

- 2 (a) Force experienced by a mass in a gravitational field **B1**
- (b) (i) total downward force on the moving pulley
- $$= 960 + (2.4 \times 9.81) = 984 \text{ N}$$
- C1**
- As the moving pulley is moving upwards at a constant speed,
net force = 0
- $$2T \cos 11.5^\circ = 984$$
- C1**
- $$T = 502 \text{ N}$$
- A1**
- (ii) Cosine of angle (with vertical) decreases
- OR angle of rope (with vertical) increase **M1**
- Hence tension increases **A1**
- (c) (i) rope / lower pulley has to be lifted up
- OR load has kinetic energy (any one) **B1**
- (ii) force applied is less than weight of load **B1**

- 3 (a) $pV/T = \text{constant}$